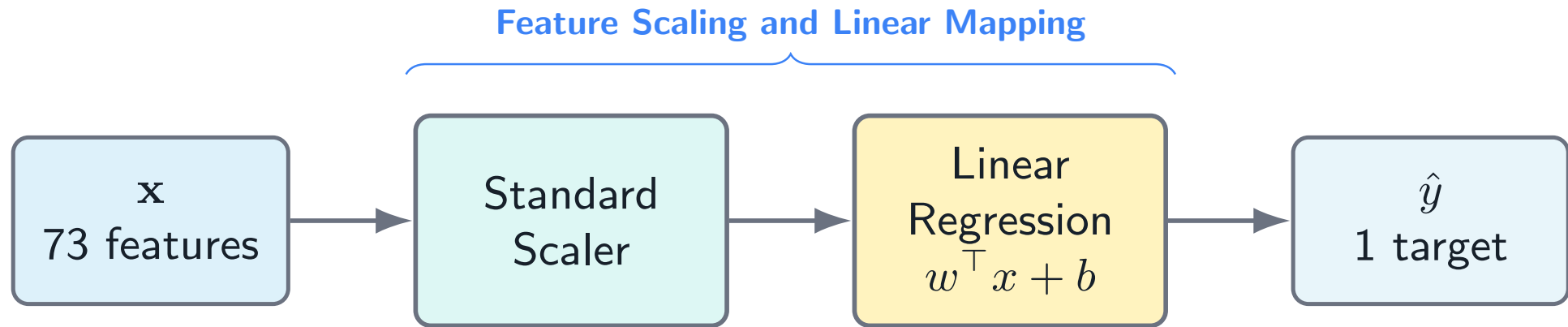


Linear Regression Architecture

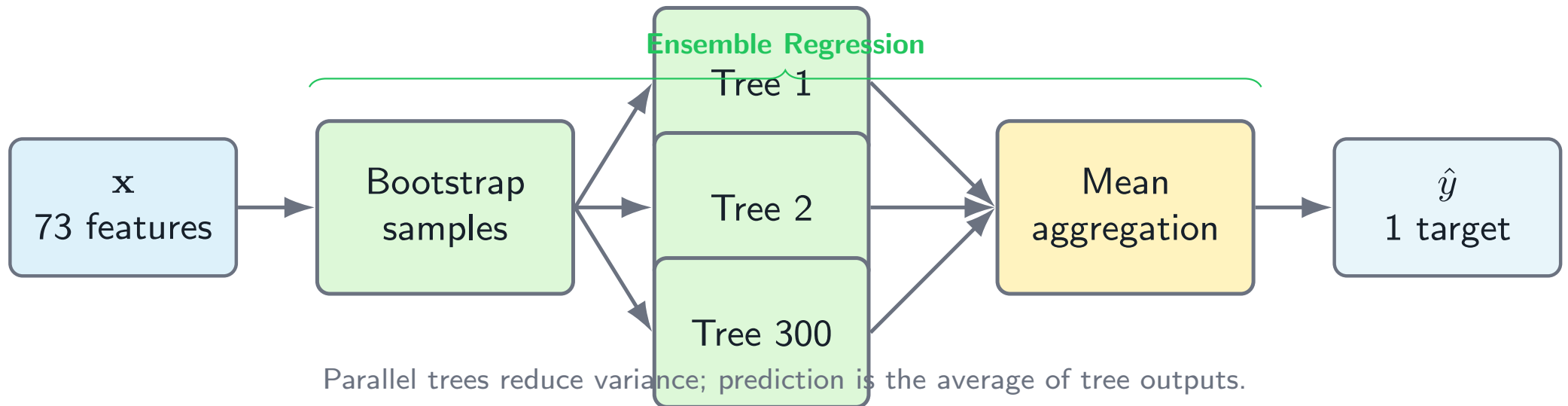
One independent regressor is trained for each target: throughput, latency, and loss.



Closed-form sklearn model, exported as one dense linear layer for edge evaluation.

Random Forest Architecture

Bagged decision-tree ensemble with averaged regression output.

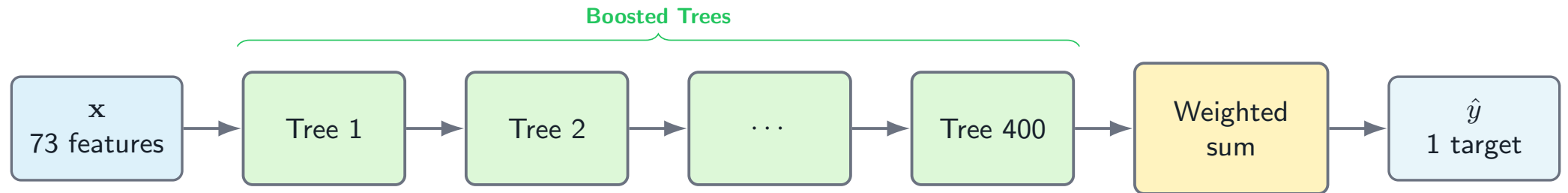


Parallel trees reduce variance; prediction is the average of tree outputs.

$n_estimators = 300$, min samples split = 4, min samples leaf = 2, max depth = unrestricted.

XGBoost Architecture

Sequential gradient-boosted decision trees with additive residual correction.

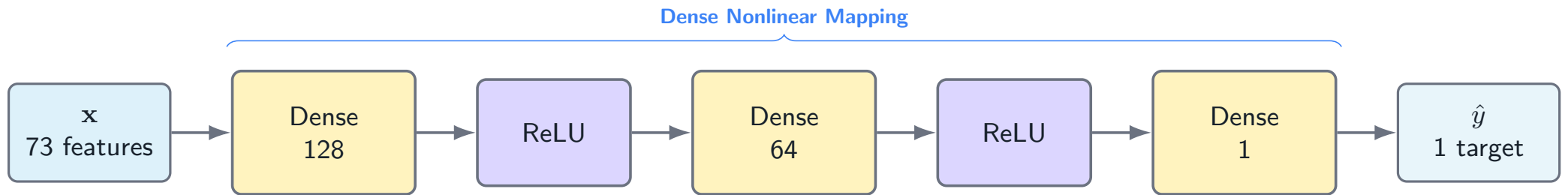


Each tree learns corrections to the current residuals.

$n_estimators = 400$, learning rate = 0.05, max depth = 8, subsample = 0.9, colsample = 0.9.

MLP Architecture

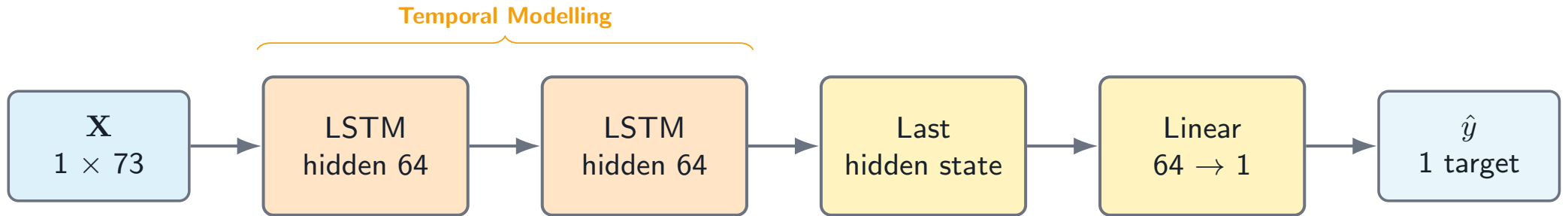
Fully connected nonlinear regressor for tabular per-station features.



Three linear layers with ReLU activations after the first two layers.

LSTM Architecture

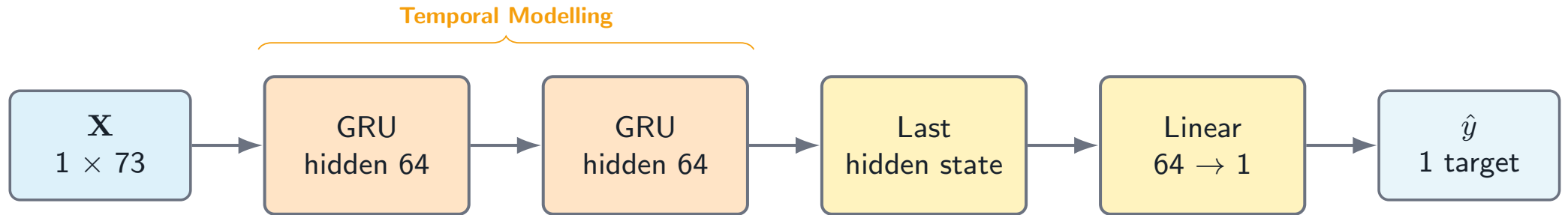
Two-layer temporal model using the last hidden state for regression.



2 recurrent layers, hidden dimension 64, batch-first sequence input.

GRU Architecture

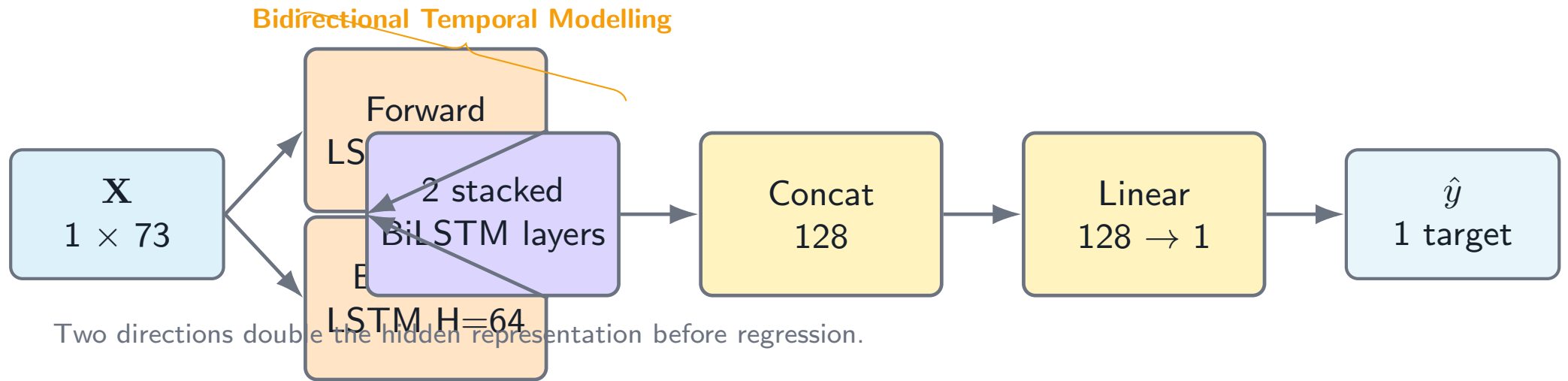
Gated recurrent model with a lighter recurrent cell than LSTM.



2 recurrent layers, hidden dimension 64, update/reset gates.

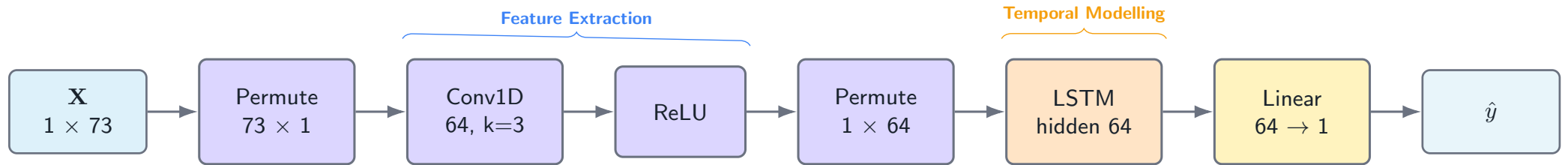
BiLSTM Architecture

Bidirectional recurrent model that concatenates forward and backward hidden states.



CNN-LSTM Architecture

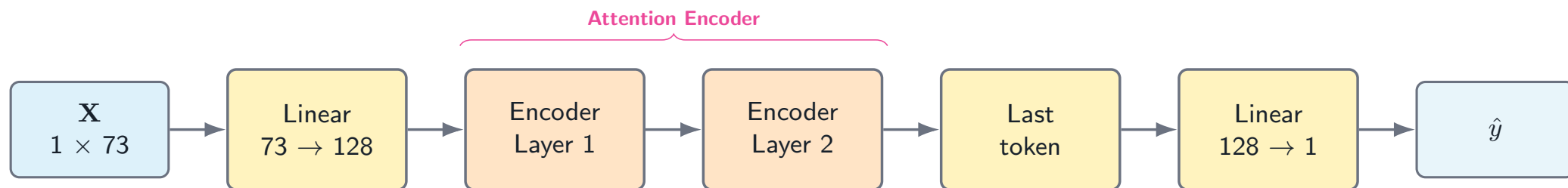
A 1D convolution extracts local feature patterns before recurrent temporal modelling.



Temporal convolution with padding, no pooling in the active model. Last recurrent output is used for the prediction head.

Transformer Architecture

Self-attention encoder for sequence-style regression over per-station features.



Two stacked TransformerEncoder layers over the embedded sequence.

Each encoder layer: 4-head self-attention, feed-forward width 512, dropout 0.1.

GNN Architecture

Graph convolution predicts from node features and topology, then pools to one output.

